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Systems and methods for actuating vehicle lights based on external light show

Abstract

A vehicle including a detection unit and a processor is disclosed. The detection unit may be configured to capture inputs associated with an external light being illuminated in proximity to the vehicle. The processor may be configured to obtain the inputs from the detection unit, and determine an illumination characteristic associated with the external light based on the inputs obtained from the detection unit. The processor may further actuate a vehicle light to illuminate based on the illumination characteristic.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8380394	12/2012	Snodgrass	701/37	B60G 17/018
9731645	12/2016	Taleb-Bendiab	N/A	N/A
10574757	12/2019	Rosen et al.	N/A	N/A
2018/0370466	12/2017	Murata	N/A	B60R 21/017
2024/0025336	12/2023	Pendikatla et al.	N/A	N/A
2024/0083374	12/2023	Ma	N/A	B60S 1/08

OTHER PUBLICATIONS

Wojciech Lewandowski, et. al., Adaptive Lighting Systems and the Method of Implementing Dynamically Adjustable On-Demand Strategies—Conclusions From Research, Technical Transactions, e2022014. <https://doi.org/10.37705/TechTrans/e2022014>, Dec. 20, 2022, pp. 1-19. cited by applicant

Walterpack, What is Dynamic Lighting?, <https://www.walterpack.com/en/what-is-dynamic-lighting/>, 2022, pp. 1-3. cited by applicant

Peter A. Kara, et. al., How I Met Your V2X Sensor Data: Analysis of Projection-Based Light Field Visualization for Vehicle-To-Everything Communication Protocols and Use Cases, MDPI Sensors 2023, 23, 1284. <https://doi.org/10.3390/s23031284>, Jan. 22, 2023, pp. 1-33. cited by applicant

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Background/Summary

FIELD

(1) The present disclosure relates to systems and methods for actuating vehicle lights based on an illumination pattern of an external light show.

BACKGROUND

(2) It is known that many users decorate their homes with lights during festivals (e.g., during Christmas) or when the users may be celebrating an event or having a get-together/party. Users typically decorate their homes in innovative ways to enhance the lighting's visual appeal and/or the home façade appearance. For example, some users decorate their parked vehicles to augment the home decoration. While such measures may assist the users to some extent, the users may desire additional means to innovatively enhance the home decoration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is set forth with reference to the accompanying drawings. The use of

the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.

(2) FIG. 1 depicts an environment in which techniques and structures for providing the systems and methods disclosed herein may be implemented.

(3) FIG. 2 depicts a block diagram of a system to control an operation of a vehicle light in accordance with the present disclosure.

(4) FIG. 3 depicts a snapshot of a vehicle projecting light on an exterior surface in accordance with the present disclosure.

(5) FIG. 4 depicts a snapshot of a trailer attached to a vehicle in accordance with the present disclosure.

(6) FIG. 5 depicts a snapshot of one or more lights connected to a vehicle power source in accordance with the present disclosure.

(7) FIG. 6 depicts a flow diagram of a method to control an operation of a vehicle light in accordance with the present disclosure.

DETAILED DESCRIPTION

(8) Overview

(9) The present disclosure describes a vehicle that may be configured to actuate one or more vehicle lights based on a light show associated with external lights that may be illuminated in proximity to the vehicle. In an exemplary aspect, the external lights may be located or disposed on adjacent to a building (e.g., a user's home), and the vehicle may be parked in proximity to the building. The vehicle may obtain images of the external lights, and may determine an illumination characteristic associated with the external lights based on the obtained images. The illumination characteristic may be, for example, an illumination duty cycle, an illumination pattern, light color(s), and/or the like, associated with the external lights. Responsive to determining the illumination characteristic, the vehicle may actuate the vehicle lights to illuminate based on the determined illumination characteristic. In some aspects, the vehicle may cause the vehicle lights to emulate the external lights, or may cause the vehicle lights to complement the external lights. In this manner, the vehicle may enable the light show associated with the external lights to extend to the vehicle lights.

(10) In further aspects, the vehicle may include one or more projectors, and may cause the projector to project a light pattern or a predefined message onto an external surface (e.g., a home wall) based on the illumination characteristic associated with the external lights. The vehicle may further include one or more speakers, and may cause the speaker to output a sound pattern that matches with the illumination characteristic associated with the external lights. In some aspects, the vehicle may be configured to customize the illumination pattern associated with the vehicle lights and/or the projector, or the sound pattern associated with the speaker based on user inputs or preferences provided by a user.

(11) The vehicle may be further configured to optimize vehicle's energy consumption while actuating the vehicle lights based on the illumination characteristic associated with the external lights. For example, the vehicle may actuate the vehicle lights when a state of charge (SoC) level associated with a vehicle battery may be greater than a predefined SoC threshold or when the vehicle may be connected or interfaced to a charger. The vehicle may further actuate only a subset of vehicle lights (and not all the vehicle lights) when the SoC level may be low. The vehicle may further actuate the vehicle lights based on a time of a day, between user-specified times, and/or ambient brightness level. The vehicle may additionally actuate the vehicle lights only when a user or another vehicle may be present in proximity to the home, and may not actuate the vehicle lights when no one may be present to view the light show.

(12) In additional aspects, the vehicle may actuate one or more trailer lights associated with a trailer attached to the vehicle based on the illumination characteristic associated with the external lights, in the same manner as the vehicle actuates the vehicle lights.

(13) The present disclosure discloses a vehicle that enables a user to extend a home-based light show to vehicle lights. The vehicle causes the vehicle lights to emulate the external lights, so that the user may decorate the user's home in an innovative manner. The vehicle further implements one or more different energy optimization measures to ensure that the vehicle's energy consumption is optimized and not unnecessarily consumed to illuminate the vehicle lights.

(14) These and other advantages of the present disclosure are provided in detail herein.

Illustrative Embodiments

(15) The disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the disclosure are shown, and not intended to be limiting.

(16) FIG. 1 depicts an example environment **100** in which techniques and structures for providing the systems and methods disclosed herein may be implemented. The environment **100** may include a vehicle **102** that may be parked or be stationary in proximity to a building **104**. In some aspects, the building **104** may be a home associated with an owner of the vehicle **102**, and the vehicle **102** may be parked outside the home. Hereinafter, in the present disclosure, the building **104** is referred to as home **104**. In other aspects, the building **104** may be a user office (or user business place) or any festival site.

(17) The vehicle **102** may take the form of any passenger or commercial vehicle such as a car, a work vehicle, a crossover vehicle, a truck, a van, a minivan, a taxi, a bus, etc. The vehicle **102** may be a manually driven vehicle, or may be configured to operate in a partially/fully autonomous mode, and may include any powertrain such as a gasoline engine, one or more electrically-actuated motor(s), a hybrid system, etc.

(18) In an exemplary aspect, the home **104** may be decorated with one or more external lights **106** (e.g., celebration or party lights) that may be disposed/placed on one or more exterior walls of the home **104**, home garden, artefacts placed at a home exterior portion, and/or the like. The homeowner may have disposed/placed the lights **106** at the home exterior portion to celebrate a special occasion, e.g., a party, a wedding, Christmas or any other festival, and/or the like. In some aspects, the lights **106** may be getting illuminated at a predefined illumination duty cycle, in a predefined illumination pattern, and/or may be emitting one or more predefined light colors. In the present disclosure, the illumination duty cycle, the illumination pattern, the light color(s), and/or other characteristics associated with the lights **106** are collectively referred to as "illumination characteristic" of the lights **106**.

(19) In some aspects, the vehicle **102** may be configured to determine (e.g., by using Artificial Intelligence/Machine Learning (AI/ML) based image processing algorithms) the illumination characteristic associated with the lights **106** based on inputs obtained from a vehicle detection unit (e.g., exterior vehicle cameras), and cause one or more vehicle lights **108** to illuminate (or actuate the vehicle lights **108**) based on the determined illumination characteristic. The vehicle lights **108** may be vehicle's interior lights and/or exterior lights. In some aspects, the vehicle **102** may cause the vehicle lights **108** to emulate and synchronize with the illumination of the lights **106** based on the determined illumination characteristic. For example, the vehicle **102** may cause the vehicle lights **108** to illuminate with the same duty cycle, in the same illumination pattern and/or in the same colors as the lights **106**. In this manner, the vehicle **102** may enable the homeowner to "extend" the home decoration or celebration lights to the vehicle **102**, thereby facilitating the homeowner to decorate the home **104** in a highly creative and innovative way. In this case, the vehicle **102** may act as a part of the home's decoration, when the home **104** may be decorated with the lights **106**.

(20) In other aspects, the vehicle **102** may cause the vehicle lights **108** to "complement" the lights

106, and may not completely emulate the illumination characteristic associated with the lights **106**. For example, in this case, the vehicle **102** may cause the vehicle lights **108** to illuminate in a first color when the lights **106** may be illuminating in a second color (which may be different from the first color), and/or may switch ON the vehicle lights **108** when the lights **106** may be dark and then switch OFF the vehicle lights **108** when the lights **106** may be illuminated.

(21) In some aspects, the vehicle **102** may cause the vehicle lights **108** to illuminate based on the illumination characteristic associated with the lights **106** when the vehicle **102** may be parked or stationary in proximity to the home **104**, and may not cause the vehicle lights **108** to illuminate when the vehicle **102** may be in motion.

(22) Further, in some aspects, the vehicle **102** may cause the vehicle lights **108** to illuminate when a count of users (e.g., a user **110**) walking in proximity to the home **104** and/or a count of vehicles (not shown) traveling past the home **104** over a predefined time duration (e.g., 1-2 minutes) may be greater than a predefined count threshold. Stated another way, the vehicle **102** may cause the vehicle lights **108** to illuminate when a rate of users and/or vehicles moving past the home **104** may be greater than a threshold, or when users/vehicles may be detected in proximity to the home **104**. In some aspects, the vehicle **102** may implement the vehicle light illumination strategy or limitation, as described above, to conserve vehicle energy that may be consumed to illuminate the vehicle lights **108**. Specifically, the vehicle **102** may conserve the vehicle energy by not illuminating the vehicle lights **108** when no one may be present in proximity to the vehicle **102** to view the illuminated vehicle lights **108**.

(23) In some aspects, the vehicle **102** may implement additional measures to conserve the vehicle energy while illuminating the vehicle lights **108** based on the illumination characteristic associated with the lights **106**. For example, the vehicle **102** may illuminate the vehicle lights **108** when the vehicle **102** may be connected with or interfaced to an Electric Vehicle (EV) charger (e.g., when the vehicle **102** may be an EV), or when a vehicle battery state of charge (SoC) may be greater than a predefined SoC threshold. If the vehicle **102** is an internal combustion engine (ICE) vehicle, the vehicle **102** may automatically remote start or auto-start when the SoC level may be low (e.g., lower than the predefined SoC threshold). The vehicle **102** may further adjust power consumption and/or illumination brightness associated with the vehicle lights **108** based on the SoC and/or battery's energy capacity. As an example, the vehicle **102** may reduce the vehicle light brightness when the SoC may be low (e.g., lower than a predefined brightness SoC threshold), and may illuminate the vehicle lights **108** at a standard or high brightness level when the SoC may be high (e.g., higher than the predefined brightness SoC threshold).

(24) In additional aspects, the vehicle **102** may illuminate only a subset of vehicle lights **108** (and not all the vehicle lights **108**) when the SoC may be low, to conserve the vehicle energy. The vehicle **102** may further control the vehicle light illumination based on a time of the day, an ambient brightness level, and/or the like. For example, the vehicle **102** may not cause the vehicle lights **108** to illuminate during afternoon time duration and/or when the ambient brightness level may be high (e.g., when it may be a sunny day), and may cause the vehicle lights **108** to illuminate during evening or night time durations. The vehicle **102** may implement additional similar power conservation measures to conserve the vehicle energy while illuminating the vehicle lights **108**. Examples of such power conservation measures are described later in the description below.

(25) Although the description above describes an aspect where the vehicle **102** causes the vehicle lights **108** to illuminate based on the illumination characteristic associated with the lights **106**, the present disclosure is not limited to such an aspect. In additional aspects, the vehicle **102** may cause one or more vehicle interior or exterior projectors (shown as projector **240** in FIG. 2) to project a light pattern or a predefined message on an exterior surface (e.g., a home wall) based on the illumination characteristic associated with the lights **106**. For example, the vehicle projector may project the light pattern or the predefined message on the home wall in the same color and/or the illumination pattern as the lights **106**.

(26) The vehicle **102** may further cause one or more vehicle speakers (shown as speaker **242** in FIG. **2**) to output sound or music in a sound pattern that may be based on (e.g., synchronized with) the illumination characteristic associated with the lights **106**. In some aspects, the vehicle **102** may additionally control/adjust the illumination pattern associated with the vehicle lights **108** and/or the output sound pattern associated with the vehicle speakers based on user inputs or preferences associated with a user (who may be the home owner or a spectator viewing the light show associated with the lights **106** and/or the vehicle lights **108**).

(27) The vehicle **102** may be further configured to control actuation and illumination of one or more trailer lights (shown as trailer lights **402** in FIG. **4**) associated with a trailer (shown as trailer **404** in FIG. **4**) that may be electrically and communicatively connected with the vehicle **102**, based on the illumination characteristic associated with the lights **106** in the similar manner as the vehicle **102** controls actuation/illumination of the vehicle lights **108**.

(28) Further vehicle details are described below in conjunction with FIG. **2**.

(29) The vehicle **102** and/or the home owner/user implement and/or perform operations, as described here in the present disclosure, in accordance with the owner manual and safety guidelines. In addition, any action taken by the home owner based on notifications provided by the vehicle **102** should comply with all the rules specific to the location and operation of the vehicle **102** (e.g., Federal, state, country, city, etc.). The notifications, as provided by the vehicle **102**, should be treated as suggestions and only followed according to any rules specific to the location and operation of the vehicle **102**.

(30) FIG. **2** depicts a block diagram of a system **200** to control an operation of the vehicle lights **108** in accordance with the present disclosure. While describing FIG. **2**, references will be made to FIGS. **3**, **4** and **5**.

(31) The system **200** may include the vehicle **102**, one or more servers **202** (or a server **202**), a user device **204** communicatively coupled with each other via one or more networks **206**. The server **202** may be part of a cloud-based computing infrastructure and may be associated with and/or include a Telematics Service Delivery Network (SDN) that provides digital data services to the vehicle **102** and other vehicles (not shown in FIG. **2**) that may be part of a vehicle fleet. In further aspects, the server **202** may provide AI/ML based image processing algorithms to the vehicle **102**, which may enable the vehicle **102** to analyze images captured by one or more vehicle interior and/or exterior cameras and perform one or more actions based on the image analysis. For example, the vehicle **102** may determine the illumination characteristic associated with the lights **106**, determine a user and/or a vehicle presence in proximity to the vehicle **102**, and/or the like based on the image analysis. The server **202** may provide the AI/ML based image processing algorithms to the vehicle **102** at a predefined frequency, or when the vehicle **102** transmits a request to the server **202** to obtain such algorithms.

(32) The user device **204** may be associated with the home/vehicle owner, and may be, for example, a mobile phone, a key fob, a computer, a laptop, tablet, a smart wearable device, or any other device with communication capabilities.

(33) The network(s) **206** illustrates an example communication infrastructure in which the connected devices discussed in various embodiments of this disclosure may communicate. The network(s) **206** may be and/or include the Internet, a private network, public network or other configuration that operates using any one or more known communication protocols such as transmission control protocol/Internet protocol (TCP/IP), Bluetooth®, Bluetooth Low Energy (BLE), Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) standard 802.11, Ultra-wideband (UWB), and cellular technologies such as Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA), High-Speed Packet Access (HSPDA), Long-Term Evolution (LTE), Global System for Mobile Communications (GSM), and Fifth Generation (5G), to name a few examples.

(34) The vehicle **102** may include a plurality of units including, but not limited to, an automotive

computer **208**, a Vehicle Control Unit (VCU) **210**, and a light control unit **212** (or unit **212**). The VCU **210** may include a plurality of Electronic Control Units (ECUs) **214** in communication with the automotive computer **208**.

(35) In some aspects, the automotive computer **208** and/or the unit **212** may be installed anywhere in the vehicle **102**, in accordance with the disclosure. Further, the automotive computer **208** may operate as a functional part of the unit **212**. The automotive computer **208** may be or include an electronic vehicle controller, having one or more processor(s) **216** and a memory **218**. Moreover, the unit **212** may be separate from the automotive computer **208** (as shown in FIG. 2) or may be integrated as part of the automotive computer **208**.

(36) The processor(s) **216** may be in communication with one or more memory devices in communication with the respective computing systems (e.g., the memory **218** and/or one or more external databases not shown in FIG. 2). The processor(s) **216** may utilize the memory **218** to store programs in code and/or to store data for performing aspects in accordance with the disclosure. The memory **218** may be a non-transitory computer-readable medium or memory storing a light control program code. The memory **218** may include any one or a combination of volatile memory elements (e.g., dynamic random-access memory (DRAM), synchronous dynamic random-access memory (SDRAM), etc.) and may include any one or more nonvolatile memory elements (e.g., erasable programmable read-only memory (EPROM), flash memory, electronically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), etc.).

(37) In accordance with some aspects, the VCU **210** may share a power bus with the automotive computer **208** and may be configured and/or programmed to coordinate the data between vehicle **102** systems, connected servers (e.g., the server(s) **202**), and other vehicles (not shown in FIG. 2) operating as part of a vehicle fleet. The VCU **210** may include or communicate with any combination of the ECUs **214**, such as a Body Control Module (BCM) **220**, an Engine Control Module (ECM) **222**, a Transmission Control Module (TCM) **224**, a telematics control unit (TCU) **226**, a Driver Assistances Technologies (DAT) controller **228**, etc. The VCU **210** may further include and/or communicate with a Vehicle Perception System (VPS) **230**, having connectivity with and/or control of one or more vehicle sensory system(s) **232** (or a “detection unit”). The vehicle sensory system **232** may include one or more vehicle sensors including, but not limited to, a Radio Detection and Ranging (radar) sensor configured for detection and localization of objects inside and outside the vehicle **102** using radio waves, sitting area buckle sensors, sitting area sensors, a Light Detecting and Ranging (lidar) sensor, door sensors, proximity sensors, temperature sensors, wheel sensors, ambient weather sensors, vehicle internal and external cameras, one or more rain sensors, capacitive moisture sensors, etc. In some aspects, the vehicle sensory system **232** may be configured to capture inputs (via the vehicle interior and exterior cameras) associated with one or more external lights that may be illuminated in proximity to the vehicle **102**. The external lights may be, for example, the lights **106** that may be located at the home **104** (as described above in conjunction with FIG. 1), and the inputs associated with the external lights may be the images of the lights **106**.

(38) In some aspects, the VCU **210** may control vehicle operational aspects and implement one or more instruction sets received from the user device **204**, from one or more instruction sets stored in the memory **218**, including instructions operational as part of the unit **212**.

(39) The TCU **226** may be configured and/or programmed to provide vehicle connectivity to wireless computing systems onboard and off board the vehicle **102** and may include a Navigation (NAV) receiver **234** for receiving and processing a GPS signal, a BLE Module (BLEM) **236**, a Wi-Fi transceiver, a UWB transceiver, and/or other wireless transceivers (not shown in FIG. 2) that may be configurable for wireless communication (including cellular communication) between the vehicle **102** and other systems (e.g., a vehicle key fob, not shown in FIG. 2), computers, and modules. The TCU **226** may be in communication with the ECUs **214** by way of a bus. The bus may be wired or wireless.

(40) The ECUs **214** may control aspects of vehicle operation and communication using inputs from human drivers, inputs from an autonomous vehicle controller, the unit **212**, and/or via wireless signal inputs received via the wireless connection(s) from other connected devices, such as the user device **204**, the server(s) **202**, among others.

(41) The BCM **220** generally includes integration of sensors, vehicle performance indicators, and variable reactors associated with vehicle systems and may include processor-based power distribution circuitry that can control functions associated with the vehicle body such as lights, windows, security, camera(s), audio system(s), speakers, wipers, door locks and access control, and various comfort controls. The BCM **220** may also operate as a gateway for bus and network interfaces to interact with remote ECUs (not shown in FIG. 2).

(42) The DAT controller **228** may provide Level-1 through Level-3 automated driving and driver assistance functionality that may include, for example, active parking assistance, vehicle backup assistance, adaptive cruise control, among other features. The DAT controller **228** may also provide aspects of user and environmental inputs usable for user authentication.

(43) In some aspects, the automotive computer **208** may connect with an infotainment system **238** (or a vehicle Human-Machine Interface (HMI)). The infotainment system **238** may include a touchscreen interface portion and may include voice recognition features, biometric identification capabilities that can identify users based on facial recognition, voice recognition, fingerprint identification, or other biological identification means. In other aspects, the infotainment system **238** may be further configured to receive user instructions via the touchscreen interface portion, and/or display notifications, navigation maps, etc. on the touchscreen interface portion.

(44) The vehicle **102** may further include one or more projectors **240** (or a projector **240**, which may be installed in a vehicle headlamp) and one or more speakers **242** (or a speaker **242**). The projector **240** may be configured to project a light pattern or a predefined message onto an external surface, e.g., a home wall associated with the home **104**. Further, the speaker **242** may be configured to output sound/music based on user inputs, and/or command signals obtained from the automotive computer **208**, the VCU **210**, and/or the unit **212**.

(45) The computing system architecture of the automotive computer **208**, the VCU **210**, and/or the unit **212** may omit certain computing modules. It should be readily understood that the computing environment depicted in FIG. 2 is an example of a possible implementation according to the present disclosure, and thus, it should not be considered limiting or exclusive.

(46) In accordance with some aspects, the unit **212** may be integrated with and/or executed as part of the ECUs **214**. The unit **212**, regardless of whether it is integrated with the automotive computer **208** or the ECUs **214**, or whether it operates as an independent computing system in the vehicle **102**, may include a transceiver **244**, a processor **246**, and a computer-readable memory **248**.

(47) The transceiver **244** may be configured to receive information/inputs from one or more external devices or systems, e.g., the user device **204**, the server(s) **202**, and/or the like via the network **206**. For example, the transceiver **244** may receive the AI/ML based image processing algorithms from the server(s) **202** via the network **206**. Further, the transceiver **244** may transmit notifications (e.g., alert/alarm signals) to the external devices or systems. In addition, the transceiver **244** may be configured to receive information/inputs from vehicle **102** components such as the infotainment system **238**, the vehicle sensory system **232**, and/or the like. Further, the transceiver **244** may transmit notifications (e.g., alert/alarm/command signals) to the vehicle **102** components such as the infotainment system **238**, the BCM **220**, the projector **240**, the speaker **242**, etc.

(48) The processor **246** and the memory **248** may be the same as or similar to the processor **216** and the memory **218**, respectively. In some aspects, the processor **246** may be an AI/ML based processor that may utilize the memory **248** to store programs in code and/or to store data for performing aspects in accordance with the disclosure. The memory **248** may be a non-transitory computer-readable medium or memory storing the light control code. In some aspects, the memory

248 may additionally store the AI/ML based image processing algorithm(s) that the vehicle **102** may obtain from the server(s) **202**.

(49) In operation, the processor **246** may obtain inputs associated with the lights **106** from the vehicle sensory system **232** (or the detection unit). The inputs may be, for example, images of the lights **106** captured by the vehicle's exterior camera. In some aspects, the processor **246** may obtain the inputs associated with the lights **106** from the vehicle sensory system **232** when the vehicle **102** may be stationary or parked in proximity to the home **104**. The processor **246** may determine that the vehicle **102** may be located at the home **104** by obtaining a real-time vehicle geolocation from the NAV receiver **234**, and matching the real-time vehicle geolocation with the home's location (that may be pre-stored in the memory **248**). Further, the processor **246** may determine that the vehicle **102** may be stationary or parked (i.e., not moving) based on inputs obtained from the VCU **210**.

(50) Responsive to obtaining the inputs associated with the lights **106** from the vehicle sensory system **232**, the processor **246** may execute the AI/ML based image processing algorithm(s) stored in the memory **248**, and determine the illumination characteristic associated with the lights **106** based on the inputs obtained from the vehicle sensory system **232**. As described above in conjunction with FIG. 1, the illumination characteristic associated with the lights **106** may include an illumination duty cycle, an illumination pattern, a light color, and/or the like associated with the lights **106**. In some aspects, the processor **246** may store the determined illumination characteristic in the memory **248**.

(51) Responsive to determining (and storing) the illumination characteristic associated with the lights **106**, the processor **246** may transmit a command signal (via the BCM **220**) to the vehicle lights **108** and actuate the vehicle lights **108** based on the determined illumination characteristic. The processor **246** may actuate the vehicle lights **108** based on the illumination characteristic associated with the lights **106** when the vehicle **102** may be parked or stationary in proximity to the home **104**, and not when the vehicle **102** may be in motion.

(52) As described above in conjunction with FIG. 1, in some aspects, the processor **246** may actuate the vehicle lights **108** such that the vehicle lights **108** emulate or synchronize the same illumination pattern/lighting behavior as the lights **106**. In this manner, the vehicle **102** may become part of the "light show" that may be getting displayed at the home **104** via the lights **106**. In other aspects, as described above in conjunction with FIG. 1, the processor **246** may actuate the vehicle lights **108** such that the vehicle lights **108** complement the lights **106**.

(53) In additional aspects, along with actuating the vehicle lights **108** as described above, the processor **246** may cause the projector **240** to project a predefined light pattern and/or a predefined message on an exterior/external surface (e.g., the home wall) based on the illumination characteristic associated with the lights **106**, as shown in FIG. 3. FIG. 3 depicts an example snapshot where the vehicle **102** projects a predefined message **302** (e.g., "Merry Christmas") on the home wall based on the illumination characteristic associated with the lights **106**. As an example, the processor **246** may cause the projector **240** to project the predefined message **302** in the same illumination pattern and/or color (or shape) as the lights **106**.

(54) In some aspects, the illumination pattern of the vehicle lights **108** and/or the projection pattern of the predefined message **302** from the projector **240** may also be customized based on user inputs or user preferences. In this case, the processor **246** may obtain, via the transceiver **244**, the user inputs/preferences from the user device **204** and/or the infotainment system **238** (or the vehicle Human-Machine Interface (HMI)). In some aspects, the processor **246** may also obtain the user inputs based on the inputs obtained from the vehicle sensory system **232** (e.g., based on user images captured by the vehicle's interior and/or exterior cameras) and/or a vehicle microphone (not shown). In an exemplary aspect, the user inputs may be in the form of a user gesture, a voice command, and/or a user input received via the vehicle HMI. As an example, the user may command the vehicle lights **108** to illuminate with a certain user-preferred duty cycle and/or color

by outputting a voice command, which may be captured by the vehicle microphone and transmitted to the processor **246**. Responsive to obtaining the user inputs as described above, the processor **246** may cause the vehicle lights **108** to illuminate and/or the projector **240** to project the predefined message **302** based on the user inputs.

(55) In further aspects, the processor **246** may cause the speaker **242** to output music or sound pattern based on the illumination pattern associated with the lights **106** and/or the obtained user inputs. As an example, the processor **246** may cause the speaker's volume to increase and decrease based on the illumination pattern of the vehicle lights **108** and/or the lights **106**, and/or cause the speaker **242** to output a song based on the user inputs. In the latter case, the user requesting the song to be played by the speaker **242** may be the homeowner or a spectator viewing the light show associated with the vehicle lights **108** and/or the lights **106**. The processor **246** may obtain the song request via the user device **204**, the infotainment system **238**, the vehicle microphone, vehicle-to-vehicle (V2V) communication, vehicle-to-infrastructure (V2I) communication, and/or the like. In additional aspects, the processor **246** may cause the vehicle lights **108** to illuminate and/or the projector **240** to project the predefined message **302** based on the melody or sound characteristics associated with the song requested by the user.

(56) In yet another aspect, the processor **246** may cause (via the BCM **220** and/or vehicle's electric power take off) the vehicle body to move up and down, or a vehicle sitting area to move up and down to match the illumination pattern associated with the lights **106** and/or the sound pattern output by the speaker **242**. Such an action may further enhance the visual appeal of the home decoration, and may enable an entertaining light show presentation for the spectators. In additional aspects, the processor **246** may match the vehicle engine's revolution per minute to the illumination characteristic associated with the lights **106** and/or the sound pattern output by the speaker **242**. In further aspects, the processor **246** may periodically activate (via the BCM **220**) snow or rain clearing measures to keep the sensors associated with the vehicle sensory system **232** and/or the vehicle lights clean/clear. In an exemplary aspect, in this case, the processor **246** may periodically activate vehicle wipers, heaters, vibration, vehicle body movement, and/or the like.

(57) Further, as described above in conjunction with FIG. **1**, the processor **246** may implement one or more measures to optimize vehicle energy consumption while illuminating the vehicle lights **108** and/or actuating the projector **240** and/or the speaker **242**. For example, the processor **246** may actuate the vehicle lights **108** based on the illumination characteristic associated with the lights **106** when the processor **246** determines that a vehicle battery SoC level may be greater than a predefined SoC threshold or when the vehicle **102** may be connected to an EV charger. Stated another way, the processor **246** may not actuate the vehicle lights **108** when the vehicle battery SoC level may be less than the predefined SoC threshold, to conserve vehicle energy. In some aspects, when the vehicle **102** does not have a high-voltage (HV) battery or the HV battery has a low SoC level (e.g., lower than the predefined SoC threshold), the user may power the vehicle lights **108** by using an infrastructure powered electric cord. In some aspects, power may also be transferred to the vehicle **102** to illuminate the vehicle lights **108** via an inductive pad installed at the home's driveway and/or an energy source from another vehicle. When the vehicle **102** may be an internal combustion engine (ICE) vehicle or a hybrid electric vehicle (HEV), the vehicle engine may be pulled up or ignited periodically to charge the vehicle battery, to enable efficient vehicle light illumination.

(58) As another example, to optimize vehicle's energy consumption, the processor **246** may illuminate only a subset of vehicle lights (and not all the vehicle lights **108**) based on the real-time battery SoC level. For example, the processor **246** may actuate all the vehicle lights **108** when the battery SoC level may be greater than a first predefined SoC threshold, and may actuate only a subset of vehicle lights when the battery SoC level may be less than the first predefined SoC threshold but greater than a second predefined SoC threshold (which may be, e.g., equivalent to the predefined SoC threshold described above). In this case, the processor **246** may not illuminate any

vehicle light when the real-time battery SoC level may be less than the second predefined SoC threshold.

(59) As yet another example, the processor **246** may adjust the vehicle light power consumption and/or a vehicle light illumination brightness based on the real-time vehicle SoC level. For example, the processor **246** may dim the vehicle lights **108** or reduce vehicle light brightness when the real-time vehicle SoC level may be low (e.g., less than a predefined brightness SoC threshold). The processor **246** may further adjust the vehicle light power consumption and/or the vehicle light illumination brightness based on a usable energy capacity level associated with the vehicle battery.

(60) As yet another example, the processor **246** may actuate/illuminate the vehicle lights **108** based on a time of day, between specific times (e.g., a user-preferred time duration of between 7 to 9 PM), and/or an ambient brightness level. For example, the processor **246** may not actuate/illuminate the vehicle lights **108** during daytime (when the ambient light may be bright), and may illuminate the vehicle lights **108** during nighttime.

(61) As yet another example, the processor **246** may illuminate the vehicle lights **108** based on the illumination characteristic associated with the lights **106** when someone may be present in proximity to the vehicle **102**/home **104** to view the light show, and may not illuminate the vehicle lights **108** when no one may be present to watch the light show. In this case, the processor **246** may determine a user and/or vehicle presence in proximity to the vehicle **102** or the home **104** (inside or outside the home **104**) based on the inputs obtained from the vehicle sensory system **232** (e.g., the vehicle's exterior cameras), home based proximity sensors, motion detectors, and/or the like. Responsive to determining the user and/or vehicle presence in proximity to the vehicle **102** or the home **104**, the processor **246** may determine a count of users walking in proximity to the vehicle **102**/home **104** and/or a count of vehicles traveling in proximity to the vehicle **102**/home **104** in a predefined time duration (e.g., 1-2 minutes). The processor **246** may actuate/illuminate the vehicle lights **108** when the count of users and/or the count of vehicles may be greater than a predefined count threshold. Stated another way, the processor **246** may illuminate the vehicle lights **108** when a rate of users/vehicles walking/traveling in proximity to the vehicle **102**/home **104** may be greater than a threshold, and may not illuminate the vehicle lights **108** when the rate may be lesser than the threshold (to conserve vehicle's energy consumption).

(62) Although the description above describes an aspect where the processor **246** actuates/illuminates the vehicle lights **108** based on the illumination characteristic associated with the lights **106**, the present disclosure is not limited to such as aspect. In additional aspects, the processor **246** may be configured to cause one or more trailer lights **402** associated with a trailer **404** attached to the vehicle **102** to actuate based on the illumination characteristic associated with the lights **106**, as shown in FIG. 4. The trailer lights **402** may be, for example, left and/or right turn signal lights, speed reduction lights, running lights, reverse lights, etc. In this manner, the processor **246** may enable the light show associated with the vehicle lights **108** and/or the lights **106** to extend to the trailer **404**, which may be connected to the vehicle **102** via a 4-pin, 7-pin, or any other electrical connector. In further aspects, the processor **246** may modulate the 12V power supplied to the trailer **404** via the vehicle battery to control any exterior or accessory holiday lights that the user may have additionally installed on the trailer **404**. The processor **246** may modulate the power supply by modulating the 12V power that may be getting transferred to the trailer connector. The use of the exterior or accessory holiday lights may enable the vehicle light show to be expanded to a larger display (for example, a side of a long tri-axle enclosed cargo trailer provides large surface area for a great light display). In some aspects, the vehicle **102** and/or the trailer **404** may be associated with the home owner. In other aspects, the vehicle **102** and/or the trailer **404** may not be associated with the home owner, but may instead be associated with a neighbor who may desire to participate in the festival lighting and may not have decorated own home.

(63) In further aspects, the processor **246** may actuate one or more additional lights **502** that may be attached to a vehicle's power source **504** (or a power pocket) based on the illumination

characteristic associated with the lights **106** and/or user preferences, as shown in FIG. **5**. In this case, the processor **246** may first obtain the user preferences associated with the illumination of the additional lights **502**. The user preferences may be associated with a time duration when to actuate each light circuit (mapped to each outlet of the power source **504**), the light pattern that may be implemented for the additional lights **502**, the relative brightness of each light, the light color, and/or the like. Responsive to obtaining the user preferences, the processor **246** may cause the additional lights **502** to illuminate based on the user preferences. In this manner, the light show may be extended to additional lights, over and above the vehicle lights **108**.

(64) FIG. **6** depicts a flow diagram of a method **600** to control an operation of the vehicle lights **108** in accordance with the present disclosure. FIG. **6** may be described with continued reference to prior figures. The following process is exemplary and not confined to the steps described hereafter. Moreover, alternative embodiments may include more or less steps than are shown or described herein and may include these steps in a different order than the order described in the following example embodiments.

(65) The method **600** starts at step **602**. At step **604**, the method **600** may include obtaining, by the processor **246**, the inputs/images associated with the lights **106** from the vehicle sensory system **232**. At step **606**, the method **600** may include determining, by the processor **246**, the illumination characteristic associated with the lights **106** based on the obtained inputs/images. At step **608**, the method **600** may include actuating/illuminating, by the processor **246**, the vehicle lights **108** based on the determined illumination characteristic associated with the lights **106**, as described above in conjunction with FIGS. **1** and **2**.

(66) The method **600** may end at step **610**.

(67) In the above disclosure, reference has been made to the accompanying drawings, which form a part hereof, which illustrate specific implementations in which the present disclosure may be practiced. It is understood that other implementations may be utilized, and structural changes may be made without departing from the scope of the present disclosure. References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a feature, structure, or characteristic is described in connection with an embodiment, one skilled in the art will recognize such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(68) Further, where appropriate, the functions described herein can be performed in one or more of hardware, software, firmware, digital components, or analog components. For example, one or more application specific integrated circuits (ASICs) can be programmed to carry out one or more of the systems and procedures described herein. Certain terms are used throughout the description and claims refer to particular system components. As one skilled in the art will appreciate, components may be referred to by different names. This document does not intend to distinguish between components that differ in name, but not function.

(69) It should also be understood that the word “example” as used herein is intended to be non-exclusionary and non-limiting in nature. More particularly, the word “example” as used herein indicates one among several examples, and it should be understood that no undue emphasis or preference is being directed to the particular example being described.

(70) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Computing devices may include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above and stored on a computer-readable medium.

(71) With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating various embodiments and should in no way be construed so as to limit the claims.

(72) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

(73) All terms used in the claims are intended to be given their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Claims

1. A vehicle comprising: a detection unit configured to capture inputs associated with an external light being illuminated in proximity to the vehicle; and a processor communicatively coupled with the detection unit, wherein the processor is configured to: obtain the inputs from the detection unit; determine an illumination characteristic associated with the external light based on the inputs obtained from the detection unit; and actuate a vehicle light based on at least a duty cycle of the illumination characteristic.
2. The vehicle of claim 1, wherein the external light is located on a wall of a user home, and wherein the processor is configured to activate the vehicle light when the vehicle is stationary or parked in proximity to the user home.
3. The vehicle of claim 1, wherein the detection unit comprises an exterior vehicle camera, and wherein the inputs associated with the external light comprise external images.
4. The vehicle of claim 1, wherein the vehicle light comprises at least one of an exterior vehicle light or an interior vehicle light.
5. The vehicle of claim 1, wherein the processor is further configured to: obtain a user input; and cause the vehicle light to illuminate based on the user input.
6. The vehicle of claim 5, wherein the user input comprises at least one of a user gesture, a user voice command, or an input received via a vehicle Human-Machine Interface (HMI).
7. The vehicle of claim 1, wherein the processor causes the vehicle light to emulate the illumination characteristic.
8. The vehicle of claim 1, wherein the processor causes the vehicle light to complement the illumination characteristic.

9. The vehicle of claim 1 further comprising a projector, wherein the processor is further configured to cause the projector to project a light pattern on an external surface based on the illumination characteristic.
 10. The vehicle of claim 1 further comprising a speaker, wherein the processor is further configured to cause the speaker to output a sound pattern based on at least one of the illumination characteristic or user inputs.
 11. The vehicle of claim 10, wherein the processor is further configured to cause a vehicle body to move up or down or a vehicle sitting area to move up or down to match the illumination characteristic.
 12. The vehicle of claim 1, wherein the processor is further configured to: determine that a vehicle state of charge (SoC) is greater than a predefined SoC threshold or the vehicle is connected or interfaced to a charger; and actuate the vehicle light to illuminate based on the illumination characteristic when the vehicle SoC is greater than the predefined SoC threshold or the vehicle is connected or interfaced to the charger.
 13. The vehicle of claim 12, wherein the processor is further configured to adjust at least one of a vehicle light power consumption or a vehicle light illumination brightness based on the vehicle SoC.
 14. The vehicle of claim 1, wherein the processor is further configured to actuate the vehicle light based on at least one of a time of a day or an ambient brightness level.
 15. The vehicle of claim 1, wherein the processor is further configured to: detect a count of users walking in proximity to the vehicle or a count of vehicles traveling in proximity to the vehicle in a predefined time duration; and actuate the vehicle light when the count of users or the count of vehicles exceeds a predefined count threshold.
 16. The vehicle of claim 1, wherein the processor is further configured to cause a trailer light associated with a trailer attached to the vehicle to actuate based on the illumination characteristic.
 17. The vehicle of claim 16, wherein the processor is further configured to modulate a power supply provided to the trailer from a battery of the vehicle, to cause the trailer light to actuate based on the illumination characteristic.
 18. The vehicle of claim 1, wherein the processor is further configured to: obtain user preferences associated with illumination of one or more lights connected to a vehicle power source; and cause the one or more lights to actuate in an illumination pattern based on the user preferences.
 19. A method to control an operation of a vehicle light, the method comprising: obtaining, by a processor, inputs associated with an external light being illuminated in proximity to a vehicle; determining, by the processor, an illumination characteristic associated with the external light based on the inputs; and actuating, by the processor, the vehicle light based on at least a duty cycle of the illumination characteristic.
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